QUANT

Number System

(Digital Root, Unit's Digit, Last two digits
& other important concepts)

Assignment 05

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Numbers Online Assignment -05

1. *J*, *K*, *L*, and *M* are four integers. The product of the larger three is 4,199, and that of the smaller three is 1,547. What is their sum?

1. 56 2. 55 3. 57 4. 53

DIRECTIONS *for questions 2 & 3:* *The numbery root of a number is defined as the number obtained by adding all the digits of a number, then adding the digits of this sum, and so on until we get a single digit number. Also, the degree of the numbery root is defined as the number of times the process of adding the digits of the number is carried out to obtain the final single digit number. e.g., the number 326 has the numbery root 2 with degree 2.*

2. Find the numbery root of 11^{208}.

1. 2 2. 6 3. 7 4. 1

3. If we write all the numbers from 237 to 738, then how many numbers have numbery root as 8?

1. 53 2. 54 3. 55 4. 56

4. For how many values of a are, a, $a + 2$, $a + 4$ prime numbers?

1. One 2. Two 3. None 4. Infinite

5. In a 4 digit number, the sum of first 2 digits is equal to that of the last 2 digits. The sum of the first and last digits is equal to the 3rd digit. Finally the sum of second and fourth digits is twice the sum of other 2 digits. What is the third digit of this number?

1. 5 2.8 3. 1 4. 4

6. Which of the following options gives a pair of co-prime numbers?

1. 125, 135 2. 33, 222 3. 25, 16 4. 61, 183

7. What will be the last two digits of the number 2^{93}?

1. 32 2. 28 3. 82 4. 92

8. What is the sum of the odd natural numbers up to 50?

1. 625 2. 650 3. 600 4. 1225

9. What are the last two digits of 7^{2008}?

1. 21 2. 61 3. 01 4. 03

10. What will be the two right most digits of the number 3^{330}?

1. 49 2. 39 3. 69 4. 89

Answer keys & Explanations

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1.	1	As the product of three integers is odd, the three integers used must be odd. Same is the case in the 2nd product of 3 integers. Now when all these would be added, the sum must be even (odd + odd + odd + odd) and only 1st option can be answer.
2.	3	$11^{208} = 2^{208}$, now make cycle of numbery roots of 2 which is 2, 4, 8, 7, 5, 1 and the cycle is of 6. Now divide 208/6 and the remainder is 4 and in the cycle at the 4th place, number is 7.
3.	4	1st number after 237 which has numbery root 8 is 242…...…251……737 and the last number is 737. Now you can find the number of terms which is 56.
4.	1	$a = 3$ is the only one value satisfying this condition. So only one value
5.	1	Let the number be *abcd*, it is given that $a + b = c + d$ -------(1) $a + d = c$ ---------(2) $b + d = 2(a + c)$ ----------(3) Solve these equations and get the number as 1854 and answer as 5. **OR** You can see choices and all the choices will fit into a number.
6.	3	1st option has 5 as a common factor, 2nd option has 3 as a common factor, 4th option has 61 as a common factor, Hence 3rd option is the answer.
7.	4	2^{93}. Using the cyclicity of 2 which is of 20, we need to find the answer. Whenever we have to find the last 2 digits of a number with base 2, we subtract 21 from the power and then we divide the power by 20 and find the remainder. So 93 – 21 = 72 divided by 20 gives 12 as remainder. So 13th number in the sequence of powers of 2 is 92. So answer is 92 which is given as 4th option.
8.	1	Here, one way of thinking is making an AP. But there is a straight way. Make the students note the points. The sum of first n odd natural numbers is n^2. Whereas the sum of first n even natural numbers is $n^2 + n$.
9.	3	On seeing the cyclicity of last 2 digits of 7, we get 07, 49, 43, 01 as the last 2 digits, so the answer is 01 which is the 3rd option.
10.	1	$3^{330.}$ We know that cyclicity of 3 is 20. ∴ We divide 330 by 20 and find the remainder which comes out to be 10. Hence 10th number in the sequence of powers of 3 is 49. So answer is 49 which is given as 1st option.